TAREA 5.3 - Desarrollo de una función factorial usando TDD

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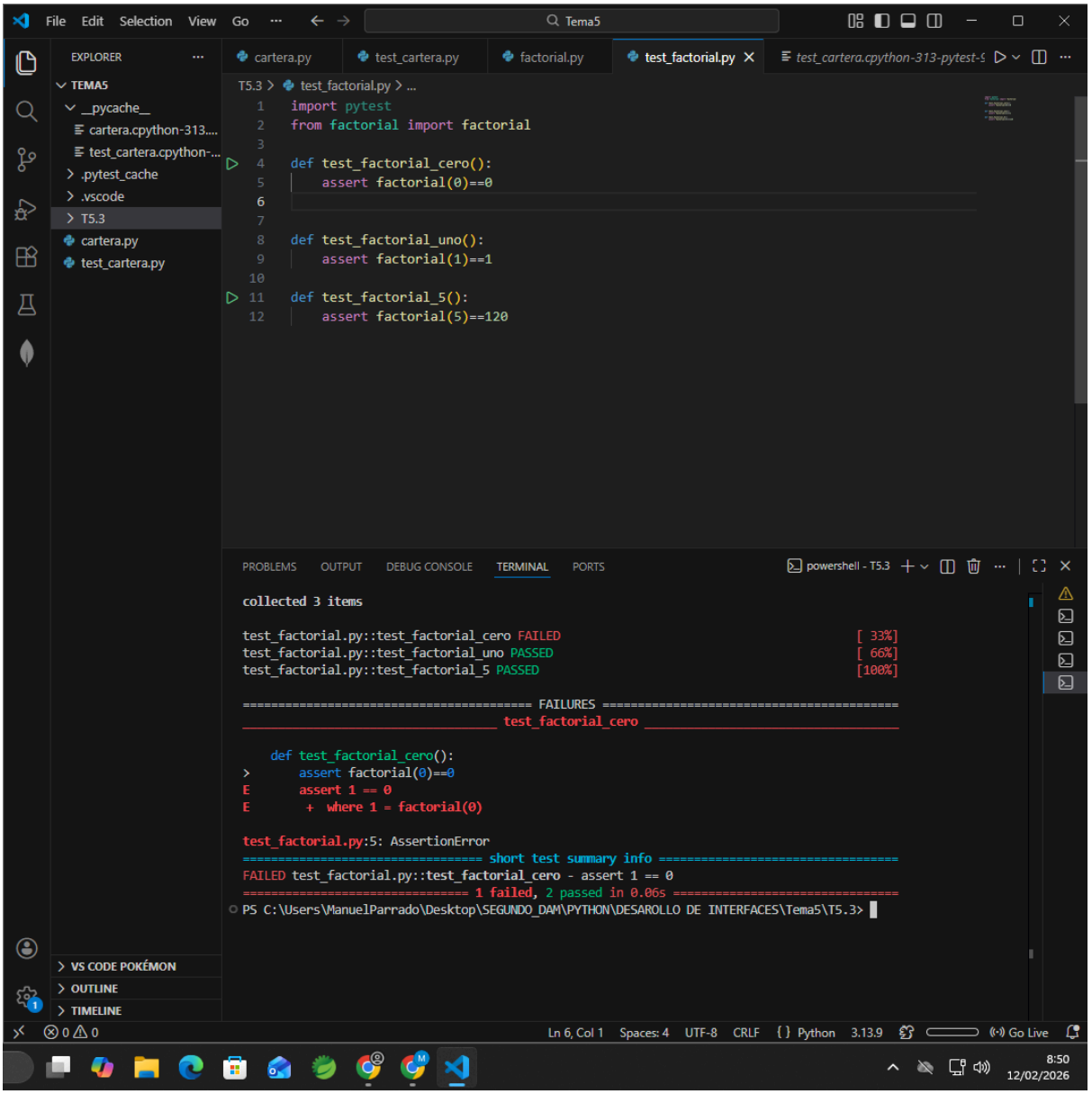
12/02/2026

Índice

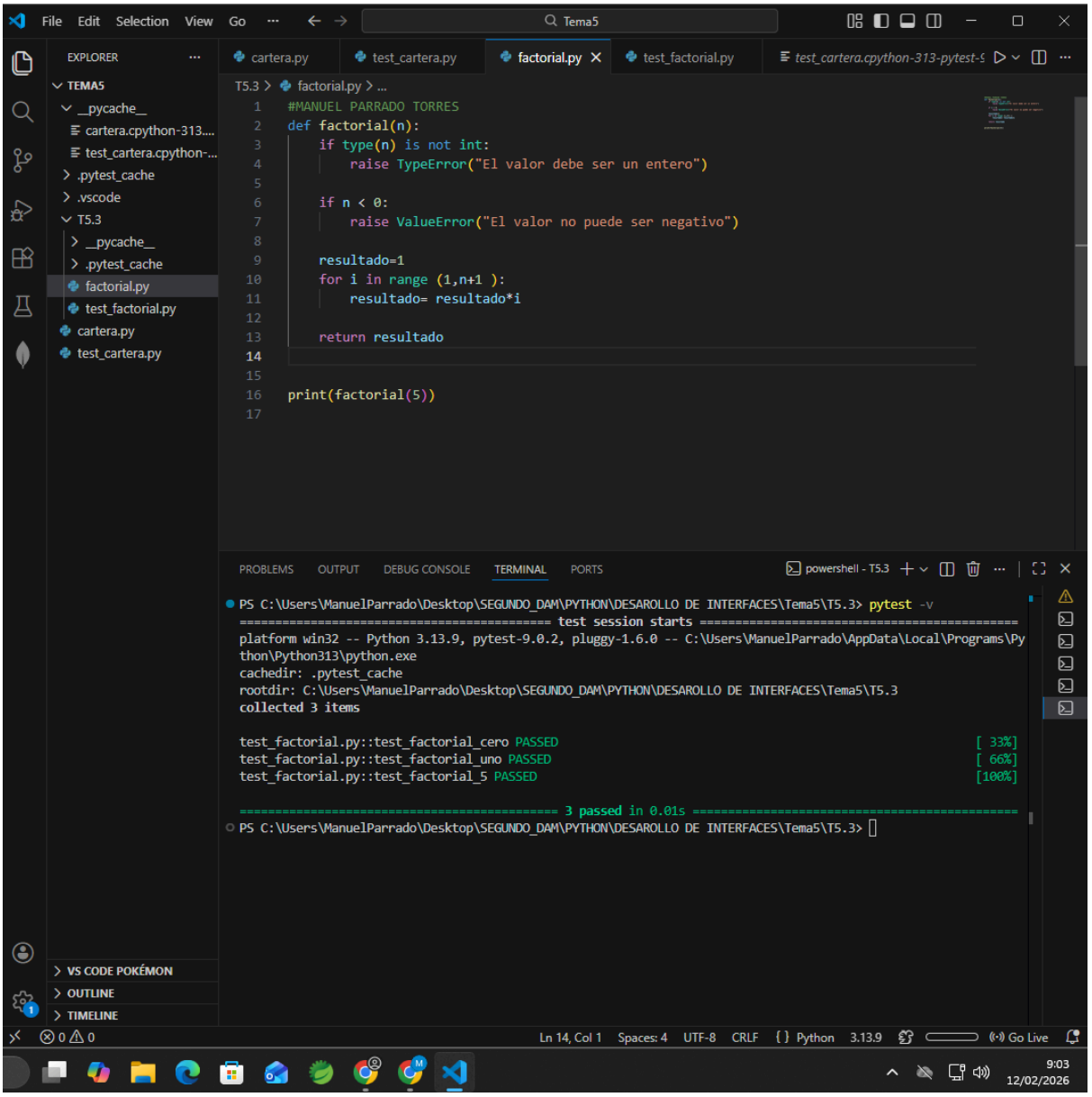
1️⃣ Fase RED – Test que falla

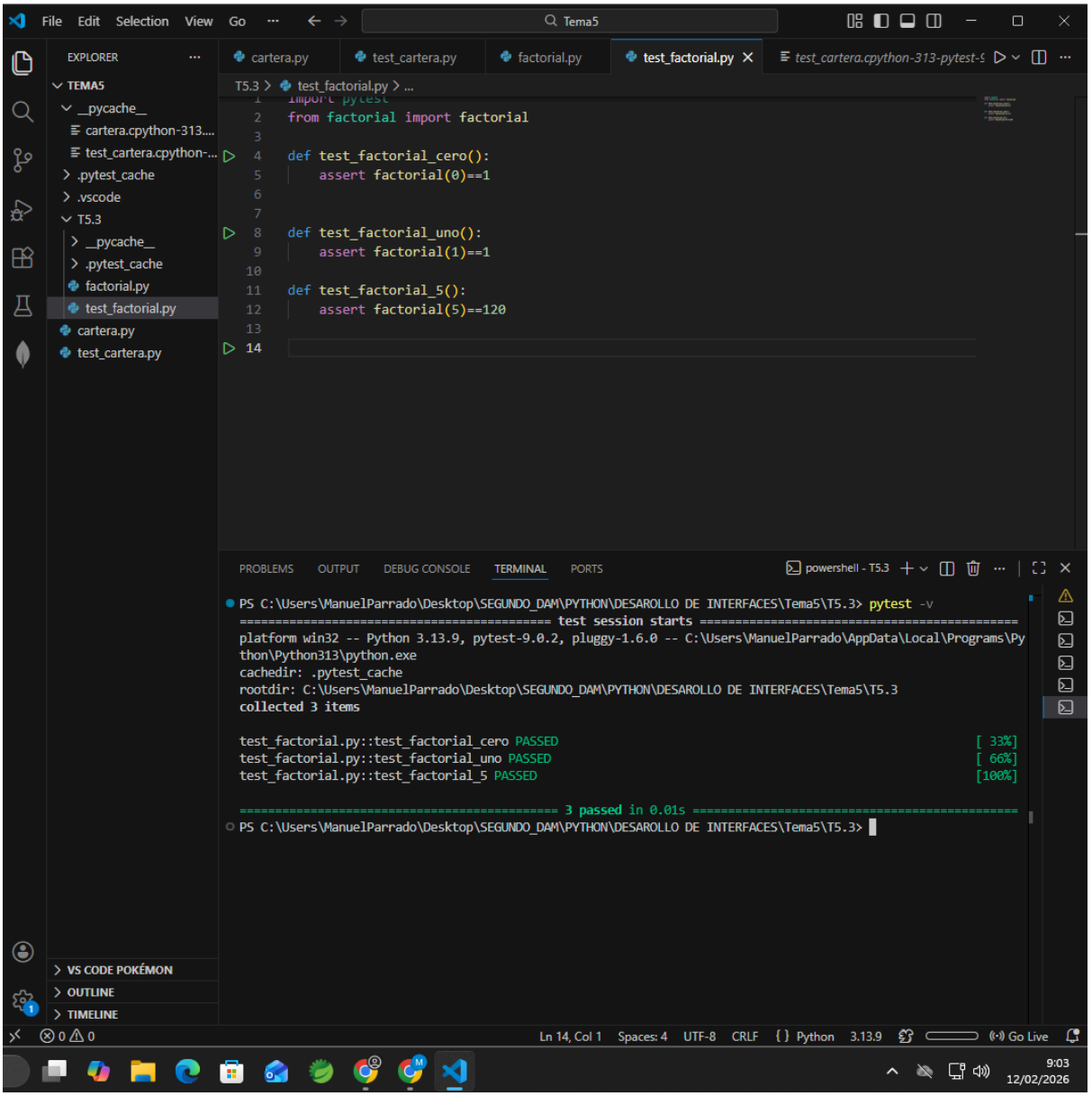
Escribe primero un test con pytest (por ejemplo, para factorial(0)).
Ejecuta pytest.
El test debe fallar.
📸 Captura obligatoria en el PDF:
Código del test
Resultado del test fallando en consola



2️⃣ Fase GREEN – Mínimo código para que pase el test

Implementa el mínimo código posible para que el test pase.
Ejecuta pytest de nuevo.
📸 Captura obligatoria en el PDF:
Código mínimo de la función
Resultado del test pasando

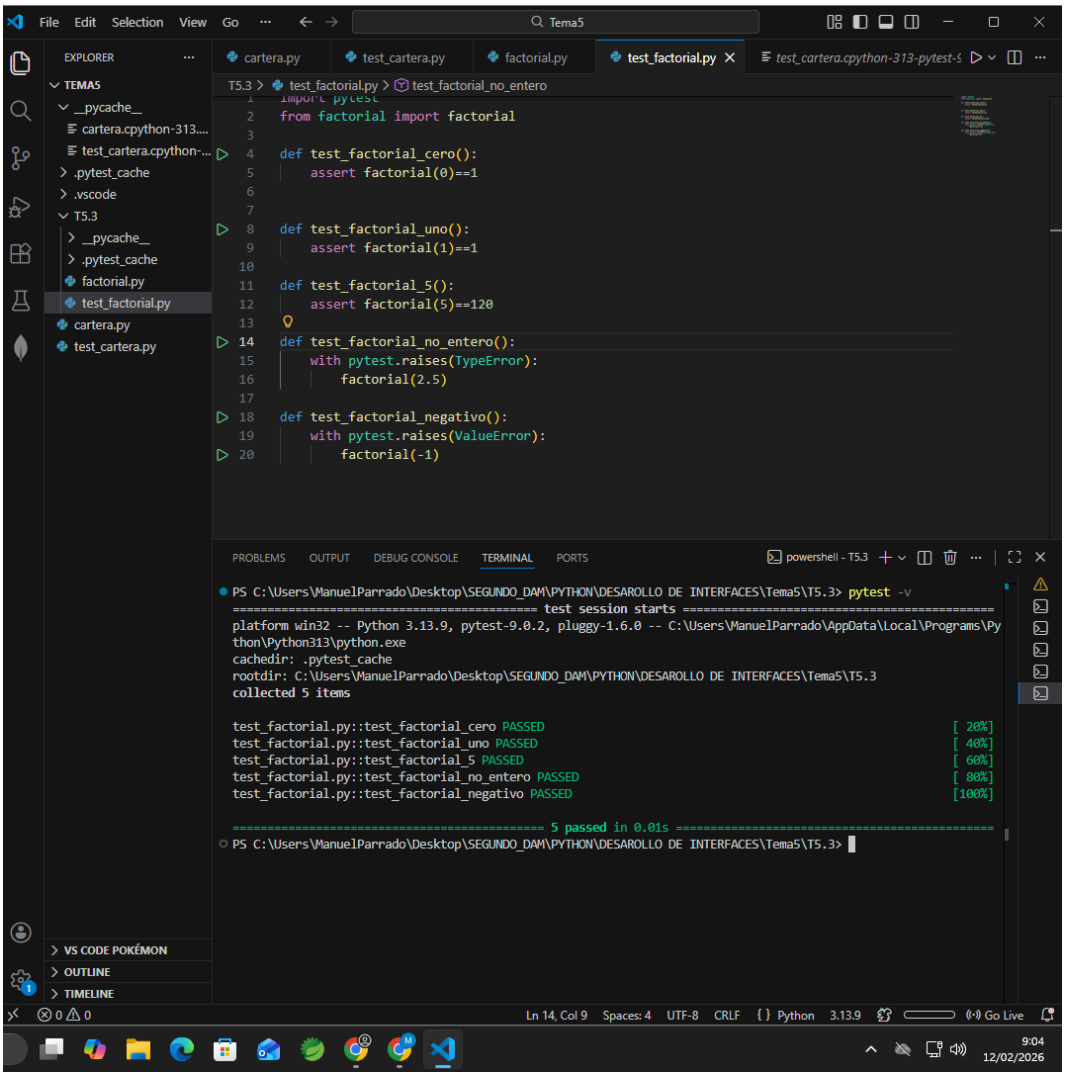


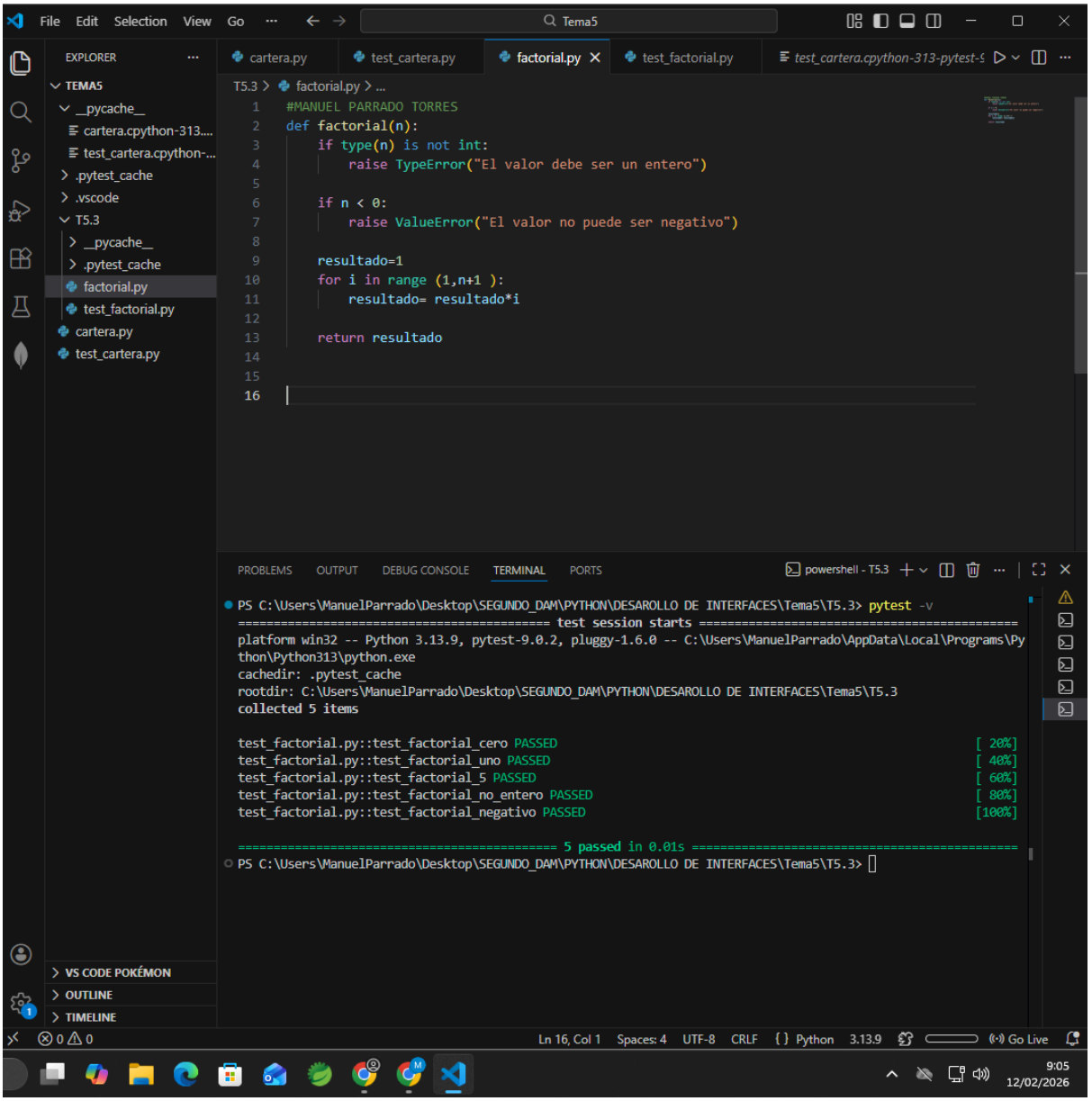

File Edit Selection View Go
Tema5
EXPLORER
TEMA5
__pycache__
cartera.cpython-313....
test_cartera.cpython-...
.pytest_cache
.vscode
T5.3
__pycache__
.pytest_cache
factorial.py
test_factorial.py
cartera.py
test_cartera.py
cartera.py
test_cartera.py
factorial.py
test_factorial.py
T5.3 > test_factorial.py > ...
import pytest
from factorial import factorial
def test_factorial_cero():
assert factorial(0)==1
def test_factorial_uno():
assert factorial(1)==1
def test_factorial_5():
assert factorial(5)==120
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
powershell - T5.3
PS C:\Users\ManuelParrado\Desktop\SEGUNDO_DAM\PYTHON\DESAROLLO DE INTERFACES\Tema5\T5.3> pytest -v
test session starts
platform win32 -- Python 3.13.9, pytest-9.0.2, pluggy-1.6.0 -- C:\Users\ManuelParrado\AppData\Local\Programs\Python\Python313\python.exe
cachedir: .pytest_cache
rootdir: C:\Users\ManuelParrado\Desktop\SEGUNDO_DAM\PYTHON\DESAROLLO DE INTERFACES\Tema5\T5.3
collected 3 items
test_factorial.py::test_factorial_cero PASSED [33%]
test_factorial.py::test_factorial_uno PASSED [66%]
test_factorial.py::test_factorial_5 PASSED [100%]
3 passed in 0.01s
PS C:\Users\ManuelParrado\Desktop\SEGUNDO_DAM\PYTHON\DESAROLLO DE INTERFACES\Tema5\T5.3>
VS CODE POKÉMON
OUTLINE
TIMELINE
Ln 14, Col 1 Spaces: 4 UTF-8 CRLF { } Python 3.13.9 Go Live
9:03
12/02/2026

3️⃣ Fase REFACTOR – Mejora del código

Añade nuevos tests de forma incremental:
factorial(1)
factorial(5)
Casos de error (negativo y no entero)
Refactoriza la función para que quede clara y general.
Los tests deben seguir pasando tras cada cambio.

📸 Capturas obligatorias en el PDF:
Nuevos tests añadidos
Código refactorizado
Ejecución final de pytest con todos los tests en verde

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factorial.py
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test_cartera.py
cartera.py
test_cartera.py
factorial.py
test_factorial.py
T5.3 > factorial.py > ...
#MANUEL PARRADO TORRES
def factorial(n):
if type(n) is not int:
raise TypeError("El valor debe ser un entero")
if n < 0:
raise ValueError("El valor no puede ser negativo")
resultado=1
for i in range (1,n+1):
resultado= resultado*i
return resultado
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
powershell - T5.3
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test session starts
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cachedir: .pytest_cache
rootdir: C:\Users\ManuelParrado\Desktop\SEGUNDO_DAM\PYTHON\DESAROLLO DE INTERFACES\Tema5\T5.3
collected 5 items
test_factorial.py::test_factorial_cero PASSED [20%]
test_factorial.py::test_factorial_uno PASSED [40%]
test_factorial.py::test_factorial_5 PASSED [60%]
test_factorial.py::test_factorial_no_entero PASSED [80%]
test_factorial.py::test_factorial_negativo PASSED [100%]
5 passed in 0.01s
PS C:\Users\ManuelParrado\Desktop\SEGUNDO_DAM\PYTHON\DESAROLLO DE INTERFACES\Tema5\T5.3>
VS CODE POKÉMON
OUTLINE
TIMELINE
Ln 16, Col 1 Spaces: 4 UTF-8 CRLF { } Python 3.13.9 Go Live
9:05
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